ASCE STANDARD

ASCE/SEI

76-23

Standard for Mitigation of Disproportionate Collapse Potential in Buildings and Other Structures

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PREFACE

Since the Ronan Point apartment building collapse in London in 1968, the structural engineering profession has engaged in serious discussion and research about causes and implications of widespread structural failure initiated by local damage. In the decades that followed, this work and additional studies of subsequent catastrophic structural failures have advanced the body of knowledge for enhancing robustness and limiting collapse when buildings are damaged. Numerous papers and several guide documents have informed structural engineers, but a single consensus perspective for mitigating disproportionate collapse has not been available.

Discussion about a standard for disproportionate collapse mitigation began when the Disproportionate Collapse Standard and Guidance Committee, met at the Structural Engineering Institute (SEI) Structures Congress in 2009. Subsequently, advocates for a standard formally proposed that SEI authorize a committee to develop a standard on the topic of disproportionate collapse mitigation. A committee was approved and met for the first time at the SEI Structures Congress in 2012. After developing a draft document and a series of ballots, the committee sent its proposed document to the American Society of Civil Engineers (ASCE) for Public Ballot in 2021. Public balloting finished in 2023, leading to the publication of this standard.

The committee endeavored to make this standard performance-based, allowing engineers maximum flexibility when finding solutions to mitigate the potential for disproportionate collapse. The standard does not dictate a course of action, nor does it prescribe solutions to resist collapse. It does not address blunders in design or construction. Guidance is given about risk assessments and avoidance, as well as characteristics that enhance collapse resistance, including the strength, ductility, deformation capacity, and robustness necessary to resist collapse without exceeding relevant limit states. These goals may be demonstrated by procedures outlined in the document, or by approved alternate rational analyses.

This standard is intended to address the same structures covered by ASCE 7-16, *Minimum Design Loads and Associated Criteria for Buildings and Other Structures*. The standard is written in mandatory language, with the expectation that it could be referenced by building codes or contracts documents.

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